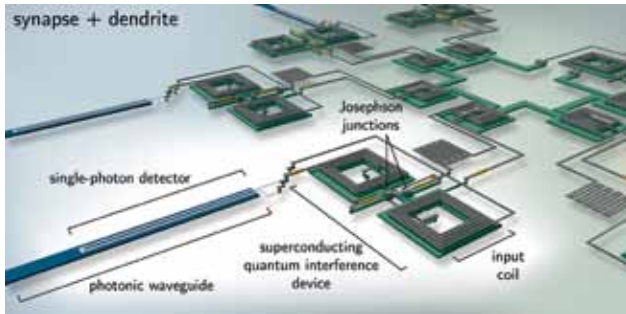




Single-photon circuit that behaves like biological synapse

Researchers at the National Institute of Standards and Technology (NIST) have designed a circuit that behaves like a biological synapse yet uses just single photons to transmit and receive signals. The circuit is based on superconducting single-photon detectors. The computation in the NIST

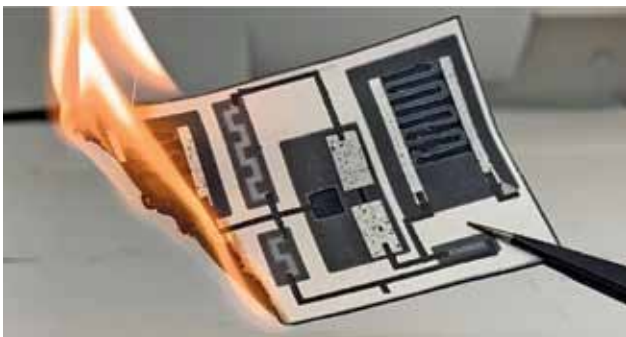


Artistic rendering of how superconducting circuits that mimic synapses (connections between neurons in the brain) might be used to create artificial optoelectronic neurons of the future (Credit: <https://www.nist.gov>)

circuit occurs where a single-photon detector meets a superconducting circuit element, a Josephson junction, which is a sandwich of superconducting materials separated by a thin insulating film. As the current exceeds a certain threshold value, the junction produces small voltage pulses called fluxons. The stored current serves as a form of short-term memory. The duration of this memory is set by the time it takes for the electric current to decay in the superconducting loops varying from nanoseconds to milliseconds, and likely beyond.

Paper based degradable electronic circuit sheet

Researchers have designed a prototype circuit using a sheet of paper with fully integrated electronic components that can be burned or left to degrade. Most circuit boards

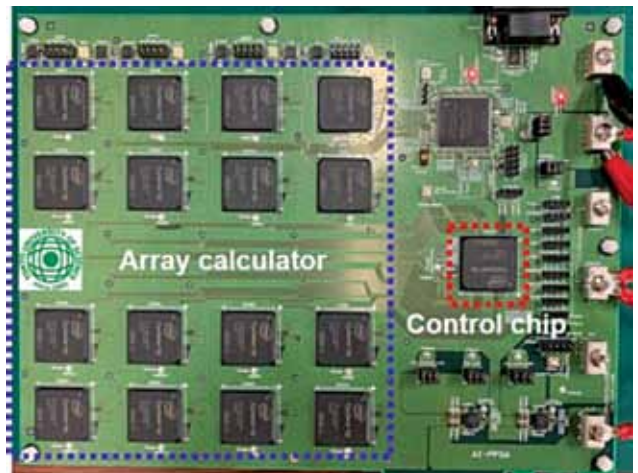


An electronic circuit printed on paper could be a more flexible and disposable option for single-use electronics (Credit: <https://techxplora.com>)

are made from glass fibres, resins, and metal wiring, and are not easy to recycle, making them undesirable for use. Researchers designed a paper based amplifier-type circuit that incorporated resistors, capacitors, and a transistor. They used wax to print channels onto a sheet of paper in a simple pattern. After melting the wax so that it soaked into the paper, the team printed semi-conductive and conductive inks, which soaked into the areas not blocked by wax. The researchers screen-printed additional conductive metal components and casted a gel based electrolyte onto the sheet. The final circuit was very flexible and thin, just like paper, and was degradable in nature.

Annealing processors that mimic quantum devices' spin

Researchers at TUS, Japan have created annealing processors that mimic the behavior of spins using quantum devices and have attempted to develop semiconductor devices using large-scale integration (LSI) technology aiming to do the same. This new method of calculation of the system's



In a new study, researchers from TUS, Japan, proposed a fully connected scalable annealing processor that, when implemented in FPGA, can easily outperform a modern CPU in solving various combinatorial optimisation problems in terms of speed and energy consumption (Credit: Takayuki Kawahara from TUS, Japan)

energy state is divided among multiple fully coupled chips first, forming an 'array calculator.' A second type of chip, called control chip, collects the results from the rest of the chips and computes the total energy, which is used to update the values of the simulated spins. They built a fully connected annealing system with 384 spins and used it to solve several optimisation problems, including a 92-node graph colouring problem and a 384-node maximum cut problem. This FPGA implementation is 584 times faster and 46 times more energy-efficient when solving the maximum cut problem.